DRUGS · HEAD AND NECK ONCOLOGY

Fluorouracil

Fluorouracil (5-FU) is a pyrimidine antimetabolite. In head and neck oncology it forms part of induction chemotherapy (TPF), older first-line palliative regimens with a platinum and cetuximab, and some chemoradiotherapy schedules. Applied topically as a 5% cream, it treats actinic keratoses and superficial basal cell carcinoma of the face and ears. Patients with dihydropyrimidine dehydrogenase (DPD) deficiency can suffer fatal toxicity, so testing before systemic treatment is now required in the UK.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

FLUOROURACIL

	SUMMARY
Class	Pyrimidine analogue antimetabolite (capecitabine is an oral prodrug)
How it works	Its metabolite FdUMP inhibits thymidylate synthase, blocking DNA synthesis; also incorporated into RNA
ENT uses	Induction TPF (docetaxel, cisplatin, 5-FU); platinum with 5-FU (with cetuximab or pembrolizumab) for recurrent or metastatic disease; some chemoradiotherapy regimens; topical 5% cream for actinic keratosis, Bowen's disease and superficial basal cell carcinoma
Dose	Commonly 750 to 1,000 mg/m2 daily as a continuous infusion for 4 to 5 days every 3 weeks (regimen-specific); topical cream once or twice daily for 2 to 4 weeks
Before treatment	DPD deficiency testing (DPYD genotyping)

2 CHAPTER TWO Using it in ENT

Practice.

2.1 Skin cancer

Topical 5-FU treats superficial lesions on the ears, nose and scalp; expect a brisk inflammatory reaction with redness, crusting and erosion before healing. It is not appropriate for nodular, infiltrative or recurrent basal cell carcinoma or for squamous cell carcinoma.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Mucositis and diarrhoea, myelosuppression, palmar-plantar erythrodysaesthesia (hand-foot syndrome), coronary vasospasm with chest pain, and severe or fatal toxicity in DPD deficiency. Topically, local inflammation and photosensitivity.

3.2 Interactions

Warfarin (marked rise in INR); brivudine and sorivudine (fatal inhibition of DPD, contraindicated); phenytoin levels rise.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Mechanism (thymidylate synthase).
- **DPD deficiency** testing.
- Topical use in skin cancer and its limits.

4.2 Common pitfalls

COMMON PITFALLS

- Using topical 5-FU for a nodular basal cell carcinoma.

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